

3. Project Proposal: Movie Recommendation System using Python

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1. Introduction / Background

In today's digital era, users have access to a vast collection of movies on various online platforms. Due to this large amount of content, it becomes difficult for users to find movies that match their interests. Recommendation systems help solve this problem by suggesting movies based on user preferences and behavior.

This project aims to develop a Movie Recommendation System using Python that suggests movies to users based on their interests using machine learning techniques.

2. Problem Statement

The problem is to design an intelligent system that can recommend movies to users based on their preferences, helping them discover relevant and interesting content easily.

3. Objectives

- To build a movie recommendation system.
 - To analyze user preferences and movie features.
 - To implement recommendation techniques such as content-based or collaborative filtering.
 - To provide personalized movie suggestions.
 - To develop a simple and user-friendly interface.
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4. Methodology

- **Data Source** – Use a publicly available dataset such as MovieLens dataset from Kaggle.
- **Data Preprocessing** – Clean and organize movie data, including titles, genres, and user ratings.
- **Feature Extraction** – Extract features such as genres, ratings, and user preferences.
- **Recommendation Model** – Implement a recommendation algorithm such as content-based filtering or collaborative filtering.
- **Evaluation** – Evaluate system performance using metrics such as accuracy and user satisfaction.
- **Prediction System** – Recommend movies to users based on their preferences and past ratings.

5. Expected Outcomes

- A working movie recommendation system.
- Personalized movie suggestions for users.
- Better understanding of recommendation algorithms.
- A simple and interactive user interface.

6. Tools and Technologies

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn
- **Techniques:** Content-Based Filtering / Collaborative Filtering
- **Dataset:** MovieLens Dataset (Kaggle)
- **Environment:** Jupyter Notebook or Google Colab
- **Deployment:** Streamlit or Flask

7. Timeline

- **Week 1:** Dataset collection and preprocessing
- **Week 2:** Feature extraction and model development
- **Week 3:** Recommendation system implementation
- **Week 4:** Testing, deployment, and report preparation

8. References

- Kaggle Dataset – <https://www.kaggle.com/>
- MovieLens Dataset – <https://grouplens.org/datasets/movielens/>
- Scikit-learn Documentation – <https://scikit-learn.org/>
- Streamlit Documentation – <https://streamlit.io/>